



Evaluation of sensitivity and specificity of routine meat inspection of Danish slaughter pigs using Latent Class Analysis

Marianne Bonde^a, Nils Toft^b, Peter T. Thomsen^a, Jan Tind Sørensen^{a,*}

^a Department of Animal Health and Bioscience, Faculty of Agricultural Sciences, University of Aarhus, Blichers Alle 20, P.O. Box 50, DK-8830 Tjele, Denmark

^b Department of Large Animal Sciences, Faculty of Life Sciences, Copenhagen University, Grønnegaardsvej 8, DK-1870 Frederiksberg C, Denmark

ARTICLE INFO

Article history:

Received 4 June 2009

Received in revised form 9 January 2010

Accepted 12 January 2010

Keywords:

Sensitivity

Specificity

Diseases

Slaughter pigs

Meat inspection

Organic

Conventional

ABSTRACT

Two groups of observers, regular meat inspectors and two veterinary researchers, respectively, conducted independent veterinary meat inspection of organs of slaughter pigs from organic or conventional production systems slaughtered at one abattoir in April 2005. A total of 3054 pigs (899 organic and 2155 conventional) were examined. The observed pathological disorders were grouped in four categories; respiratory disorders (RESP), parasitic disorders (PAR), intestinal disorders (INT) and heart disorders (HEART). Using a latent class model, the sensitivity (Se) and specificity (Sp) of meat inspectors and researchers were estimated without the assumption of an available gold standard. The Se for the regular meat inspection was found to be 0.92, 0.16, 0.24 and 0.49 for RESP, PAR, INT and HEART, respectively. The Se for the researchers was found to be 0.81, 0.96, 0.91 and 0.68 for RESP, PAR, INT and HEART, respectively. The Sp for the regular meat inspection was found to be between 0.98 and 1.00 for the four categories. The Sp for the researchers was found to be 0.83, 0.94, 0.76 and 0.99 for RESP, PAR, INT and HEART, respectively. The low Se of ordinary meat inspection for parasitic, intestinal and heart disorders will cause a significant underestimation of the prevalence of diseases reported to the pig producers. Based on our results the true prevalence of diseases (conventional vs. organic slaughter pigs) was (in %): 42 vs. 16, 5 vs. 51, 5 vs. 12 and 9 vs. 5 for RESP, PAR, INT and HEART, respectively.

© 2010 Elsevier B.V. All rights reserved.

1. Introduction

All pigs slaughtered in Denmark are subject to a routine meat inspection at the slaughterhouses (ante mortem and post mortem examination). The purposes of this meat inspection are dual: To secure safe products for human consumption, and to evaluate animal health. To be able to achieve these goals it is imperative that the meat inspection has acceptable sensitivity and specificity. A high sensitivity is needed in order to be able to detect most disorders that may potentially harm the welfare of the pigs or in some circumstances be a human health hazard. A high

specificity, on the other hand, is needed in order not to unjustly put economic or legal punishments on farmers sending truly healthy pigs for slaughter. Both a low sensitivity and a low specificity will have negative effects. Whether a low sensitivity or a low specificity is considered most unfavorable depends on a specific evaluation of the potential negative effects. That is, whether missing some disorders (thereby perhaps compromising animal welfare and/or human health) or 'punishing' some farmers unjustly is considered most unfavorable.

Sensitivity and specificity of routine meat inspection examinations of pigs have only been evaluated in a few studies so far. Enøe et al. (2003) evaluated sensitivity and specificity of routine meat inspection of chronic pleuritis in 4 Danish slaughterhouses. They found an increasing sensitivity from 1993–1994 to 1997–1998. Davies et al.

* Corresponding author. Tel.: +45 89 99 13 43; fax: +45 89 99 15 25.
E-mail address: Jantind.Sorensen@agrsci.dk (J.T. Sørensen).

(1996) generally found acceptable agreement, based on Kappa-statistics, among two groups of veterinarians (10 and 11, respectively) evaluating pneumonia, pleuritis, white spots on livers and nasal turbinate atrophy.

Animal disease information based on routine meat inspections of pigs is generally communicated with reference to crude estimates of apparent prevalence, not taking sensitivity and specificity into account. However, even in the absence of a gold standard, sensitivity and specificity of the meat inspection and true prevalence can be calculated using latent class models on the condition that we have at least two independent evaluators and two populations with differences in true prevalence (Hui and Walter, 1980).

In a larger Danish study regarding health disorders in slaughter pigs, routine meat inspection data, including 21,506 pigs from 16 organic pig producers and 217,051 pigs from 52 conventional pig producers, were analyzed (Bonde et al., 2006). All pigs were slaughtered during 2004, at the same slaughter plant and with similar carcass weight. The apparent prevalences (%) for conventional vs. organic pigs were: 27.9 vs. 11.6 for respiratory disorders, 0.7 vs. 8.2 for parasitic disorders, 1.4 vs. 0.8 for intestinal disorders, and 2.3 vs. 2.5 for heart disorders. In a German study, Ebke and Sundrum (2004) also found differences between organic and conventional slaughter pigs in the apparent prevalence of lung diseases and milky spots in the liver. Thus, it appears as if the organic and conventional pigs could constitute different populations with respect to disease prevalence.

The objective of the present study was to evaluate sensitivity and specificity of the current Danish meat inspection of slaughter pigs for different disease complexes/manifestations using a latent class model. A secondary objective was to estimate (and compare) the true prevalence of certain disease complexes in organic and conventional slaughter pigs.

2. Materials and methods

2.1. Data collection and management

Data were recorded at a public Danish slaughterhouse (Grindsted), slaughtering approximately 400 pigs per hour, during the days 11th, 13th, 15th, and 18th of April 2005. At these days, the organs of all organic and conventional pigs that were slaughtered within 2×1 hour's slaughtering, between 9.00 and 14.30, were examined by one of two researchers from the Faculty of Agricultural Sciences, in addition to the routine meat inspection. The two researchers were veterinarians with previous professional experience in routine meat inspection. The persons performing the routine meat inspection were not informed about the purpose of the presence of the two researchers and therefore were expected to have performed their meat inspection as they usually did. A total of 3054 pigs (899 organic and 2155 conventional) were examined. Due to practical limitations, only the intestines (stomach, small and large intestines) and the plucks (lungs, heart, diaphragm, liver and kidneys) were examined by the researchers. One of the researchers examined the intestines

of all animals immediately after the ordinary meat inspection of the organs, while the other researcher examined the plucks prior to the routine inspection of these organs. The carcasses of the pigs were only examined by the routine inspection, and pathological findings related to the carcass were thus not considered in this study.

All observed disorders were recorded using the standard protocol for routine meat control (by researchers as well as meat inspectors). For the remainder of this analysis, researchers and meat inspectors were considered as two different evaluators, regardless of whether several meat inspectors contributed to the recordings. Afterwards, the disease recordings were grouped into four organ categories for the analysis. The categories used for the analysis were the disease complexes occurring in each of the organs examined, i.e. intestines, lungs, heart and liver; intestinal disorders (INT), e.g. peritonitis and enteritis (acute or chronic); respiratory disorders (RESP), e.g. pneumonia and pleuritis (acute or chronic); heart disorders (HEART), e.g. pericarditis; and parasitic disorders (PAR), e.g. liver cirrhosis or milky spots in liver caused by migrating parasites. These disease complexes may be caused by a variety of different infectious agents (bacteria, virus and parasites). For the analysis 'a disease' was defined as macroscopic pathological changes irrespective of the presence or absence of a causative agent as this was not evaluated.

2.2. Statistical analyses

The data were analyzed using the latent class model suggested by Hui and Walter (1980). However, we adopted a Bayesian approach to avoid the complications of the maximum likelihood estimation imposed by the combination of a relatively small sample size (Toft et al., 2005) and the expected relatively low prevalence (and difference between populations) for some of the diseases. Essentially, the Hui-Walter model estimates the sensitivity (Se) and specificity (Sp) of two (or more) tests/evaluators as well as the prevalence (p) in two (or more) populations under the following assumptions: the tests/evaluators are conditionally independent given disease/infection status; the properties of each test/evaluators are constant across the populations, and the prevalence of the true (latent) infection status differ between populations, see e.g. Branscum et al. (2005) for further discussion of Bayesian modeling of latent class models for estimation of Se and Sp.

In a Bayesian analysis all parameters are modelled using distributions. Hence, prior distributions for the diagnostic capabilities of the evaluators and the prevalence within the organic and conventional populations must be specified. For this analysis, no relevant prior information about the evaluators and the prevalences were available. Thus, we have chosen to use uninformative priors in the shape of uniform distributions on the interval between zero and one, modelled using the Beta(1,1) distribution.

The model was implemented in OpenBUGS (Thomas et al., 2006) through BRugs in R (R Development Core Team, 2008). OpenBUGS uses a Markov Chain Monte Carlo (MCMC) sampling algorithm to obtain a Monte Carlo (MC) sample from the posterior distribution. The first 5000

Table 1

Data used for the analyses, collected during routine meat inspection (meat inspector) and a special meat inspection by two researchers (researcher) at a Danish slaughterhouse. Data are shown for each of the disease complexes intestinal disorders (INT), heart disorders (HEART), respiratory disorders (RESP) and parasitic disorders (PAR).

Disease complex	Type of pigs	Researcher/meat inspector			
		++	+–	–+	– –
INT	Conventional	19	579	6	1551
	Organic	22	271	2	604
HEART	Conventional	54	87	49	1965
	Organic	22	15	13	849
RESP	Conventional	676	268	179	1032
	Organic	111	137	38	613
PAR	Conventional	15	203	0	1937
	Organic	68	395	3	433

MC samples were discarded as a burn-in to allow convergence and the following 10,000 iterations were used for posterior inference. Convergence of the MCMC chain after the initial burn-in was assessed by visual inspection of the time-series plots of selected variables as well as Gelman-Rubin diagnostic plots using three sample chains with different initial values as suggested in Toft et al. (2007a).

Posterior inference was done by calculating means and 95% posterior credibility intervals (PCI, the Bayesian analog of a confidence interval) of the prevalence in the organic and conventional populations, and the Se and Sp of the researcher and meat inspector, respectively, for each of the disease complexes under investigation. Furthermore, to examine the differences in disease prevalences, the Bayesian posterior probability (POPR) for the expression $p_{\text{organic}} < p_{\text{conventional}}$ (or the reverse depending on disease) was calculated as the proportion of MC samples for which the expression was true. The POPR may be interpreted similarly to a p -value of the underlying hypothesis in traditional frequentist statistics (Toft et al., 2007b).

3. Results

The tabulated data of the researcher vs. meat inspection findings for the four different disease complexes stratified by organic/conventional pigs are given in Table 1. Based on these data, the estimated Se and Sp for the two types of evaluator as well as prevalences among organic and conventional pigs for each of the four disease complexes are given in Table 2. As expected, there were substantial differences in the different disease prevalences for organic and conventional pigs, respectively. All of the differences could be considered highly significant with a POPR < 0.001.

Table 2

The estimated Se and Sp of the researcher and meat inspector, respectively, for the four different disease complexes as well as the prevalence among organic (p_{organic}) and conventional (p_{conv}) pigs (95% credibility intervals in brackets).

	Prevalence (%)		Researcher		Meat inspector	
	p_{conv}	p_{organic}	Se	Sp	Se	Sp
INT	5 [1; 13]	12 [4; 22]	0.91 [0.75; 1.00]	0.76 [0.71; 0.82]	0.24 [0.21; 0.62]	1.00 [0.99; 1.00]
HEART	9 [6; 12]	5 [3; 8]	0.68 [0.51; 0.95]	0.99 [0.97; 1.00]	0.49 [0.38; 0.71]	0.99 [0.98; 1.00]
RESP	42 [37; 46]	16 [12; 20]	0.81 [0.77; 0.86]	0.83 [0.80; 0.86]	0.92 [0.84; 0.99]	0.98 [0.95; 1.00]
PAR	5 [3; 8]	51 [46; 55]	0.96 [0.91; 0.99]	0.94 [0.92; 0.97]	0.16 [0.12; 0.19]	1.00 [1.00; 1.00]

INT: intestinal disorders, HEART: heart disorders, RESP: respiratory disorders, PAR: parasitic disorders.

In conventional pig production the dominant disease complex is RESP; true prevalence 42%, whereas in organic pig production, PAR, is the dominant disease complex with a true prevalence of 51%. The overall pattern of the properties of the evaluators suggests that the researchers were more sensitive than the meat inspectors, but at the cost of a decrease in specificity. The sensitivity for the meat inspectors was particularly low for PAR (0.16; 95% PCI: 0.12–0.19) and for INT (0.24; 95% PCI: 0.21–0.62).

4. Discussion

Using latent class analyses we were able to estimate the diagnostic properties of meat inspection at slaughter compared to a team of researchers without assuming either were a gold standard.

The overall impression regarding the evaluation of diagnostic properties was that the meat inspection had a low sensitivity compared to the researchers but a very high specificity. This could indicate that there were different thresholds for defining a positive case between the researchers and the meat inspectors, where the latter require more 'typical' clinical signs in order to declare a pig positive, thus reducing the false-positives to a minimum. However, this pattern was not unambiguous. For heart disorders, the reduction in Sp for the researchers appeared to be negligible compared to the improvement in Se. For respiratory disorders, the researchers had a lower Se as well as Sp. For practical reasons the researchers did not examine the body of the carcass. This might marginally underreport respiratory disorders compared to the meat inspectors as some cases of pleuritis are more easily seen on the wall of the thorax. Enøe et al. (2003) found a Se in the range 29–87% for chronic pleuritis at a traditional meat

inspection and a Sp of 93.3–99.6%. For the disease complexes INT and PAR, there was a decrease in Sp for the researchers compared to the meat inspectors, potentially resulting in (too) many false-positives. However, there was also a 4–5 times improvement in the mean Se. Particularly the Se of the meat inspectors for the disease complex INT was subject to a substantial uncertainty due to the relatively low prevalence of this disease complex. Still, the estimates of Se for the meat inspectors for both disease complexes are less acceptable, when compared to the sensitivities achieved by the researchers. On the other hand, a Sp of 76% might also be unacceptable. Thus an optimal threshold probably lies somewhere in between the two groups of evaluators. The study design did not allow the possibility to estimate Se and Sp for individual meat inspectors or researchers.

From a practical veterinary perspective the results from this study may be useful when using meat inspection data for an evaluation of herd health status. Meat inspectors generally had a high Sp for all disease complexes and an acceptable Se for RESP only. This means that the individual farmer and his/her veterinarian may be relatively confident that a report of one or more cases from the meat inspection also reflects a true presence of disease (due to the high Sp). On the other hand, no reporting of cases from the meat inspection may not guarantee that the pigs slaughtered were truly healthy (due to the low Se).

The approach chosen for these analyses simultaneously estimated the prevalences and diagnostic properties. It was demonstrated in Vacek (1985) that ignoring a potential conditional dependence between tests/evaluators could lead to bias in the estimates of test/evaluator properties. Toft et al. (2003) demonstrated how adding or removing tests changed the prevalence in a population, something which can only be seen as a change in disease definition. The conditional independence of the researcher vs. meat inspector can be questioned, however, as discussed in Toft et al. (2005) including conditional dependence given disease status in the two test/evaluators, two population scenario introduces more problems than it solves. There are different means of addressing the problem; however, it requires either the use of more explicit prior information or data containing more evaluators, i.e. three or more sets of different evaluators simultaneously evaluating the same carcasses. For the present study, it suffices to acknowledge that there is some correlation between the sets of parameters and that the disease in question should be interpreted more like 'presence of visual indications' rather than the disease itself. Whether or not this distinction is important is in some way dependent on the specific disease complex as well as the overall purpose of the inspection. Often a detailed pathological examination has been used as a classical gold standard for estimating Se and Sp of meat inspections. However, even an extended pathological examination would not be a perfect gold standard (Enøe et al., 2003). Although for the disease definition of this study, it would be close to a perfect test. Both observer groups in our study used the general applied meat inspection protocol and both groups made the inspection in a realistic meat inspection

setting; i.e. we were in particular looking at the practical potential for improving the sensitivity of a normal meat inspection.

The organic pigs had a higher prevalence of the disease complexes PAR and INT, whereas the conventional pigs had a higher prevalence of REP and HEART. The disease prevalences reported in the current study are based on a relatively small sample of pigs from four days of slaughter and without information on herd origin, and must be interpreted with caution. However, if we compare the results in this study with the apparent prevalences found in a larger study on similar populations reported by Bonde et al. (2006) the differences in RESP and PAR between the two production systems are consistent. Comparable differences between the two systems were also observed by Ebke and Sundrum (2004), although the differences between the two systems seem to be smaller in Germany. In an Austrian study the pathological findings of pigs from 48 organic pig herds were recorded at slaughter (Baumgartner et al., 2003). They reported that 50% of the slaughter pigs displayed milky spots in the liver and 24% had pneumonic lesions. These levels were similar to the true prevalences of PAR and RESP estimated from this study. As to whether or not the potential differences in, e.g. pathogens between organic and conventional farms would manifest as differences in the Se and Sp for the organic and conventional pigs, respectively, and thus violate the model assumptions is not possible to establish in the current study, nor in any of the available literature.

5. Conclusion

Compared to the researcher evaluation, the regular meat inspection in general lacked sensitivity. This will underestimate the prevalence, which is reported back to the pig producers. It seems to be possible to improve the sensitivity for some disease complexes without a marked decrease in specificity. The true prevalence of the diseases was found to be high, especially for PAR in organic pigs and for RESP in conventional pigs.

Conflict of interest statement

The authors declare that they have no conflicts of interest regarding the present study and manuscript.

Acknowledgments

The study was funded by Svineafgiftsfonden, Fonden for Økologisk Landbrug and Dept. of Animal Health, Welfare and Nutrition. We acknowledge the contribution from Niels Peter Baadsgaard in the data collection phase and by Lene Hegelund in the editing of the data.

References

- Baumgartner, J., Leeb, T., Gruber, T., Tiefenbacher, R., 2003. Husbandry and animal health on organic pig farms in Austria. *Anim. Welfare* 12, 631–635.
- Bonde, M., Hegelund, L., Sørensen, J.T., 2006. Forskellige sundhedsproblemer hos økologiske henholdsvis konventionelle indendørs

- slagtesvin. (Differences in the health problems of organic and conventional fattening pigs). Forskningsnytt om økologisk landbrug i Norden 1, 8–9 (in Danish).
- Branscum, A.J., Gardner, I.A., Johnson, W.O., 2005. Estimation of diagnostic-test sensitivity and specificity through Bayesian modeling. *Prev. Vet. Med.* 68, 145–163.
- Davies, P.R., Bahnson, P.B., Grass, J.J., Marsh, W.E., Dial, G.D., 1996. Agreement among veterinarians evaluating gross lesions of lungs, livers, and nasal turbinates of pigs. *J. Am. Vet. Med. Assoc.* 209 (4), 823–826.
- Ebke, M., Sundrum, A., 2004. Qualitätssicherung in der ökologischen Schweinemast [Ensuring quality in organic pig production]. Paper presented at 8. Wissenschaftstagung Ökologischer Landbau, Kassel, 1. –4. März 2005; Published in Heß, J and Rahmann, G, Eds. Ende der Nische, Beiträge zur 8. Wissenschaftstagung Ökologischer Landbau. Kassel University Press GmbH, Kassel, 337–340.
- Enøe, C., Christensen, G., Andersen, S., Willeberg, P., 2003. The need for built-in validation of surveillance data so that changes in diagnostic performance of post-mortem meat inspection can be detected. *Prev. Vet. Med.* 57 (3), 117–125.
- Hui, S.L., Walter, S.D., 1980. Estimating the error rates of diagnostic tests. *Biometrics* 36, 167–171.
- R Development Core Team, 2008. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria ISBN 3-900051-07-0, <http://www.r-project.org/>
- Thomas, A., O'Hara, B., Ligges, U., Sturtz, S., 2006. Making BUGS Open. *R News* 6, 12–17.
- Toft, N., Nielsen, L.R., Højsgaard, S., 2003. Exploring disease definitions of Salmonella Dublin in latent class analysis. In: Proceedings of the 10th International Symposium on Veterinary Epidemiology and Economics, Vina del Mar, Chile. November 17–21.
- Toft, N., Jørgensen, E., Højsgaard, S., 2005. Diagnosing diagnostic tests: evaluating the assumptions underlying the estimation of sensitivity and specificity in the absence of a gold standard. *Prev. Vet. Med.* 68, 19–33.
- Toft, N., Innocent, G.T., Gettinby, G., Reid, S.W.J., 2007a. Assessing the convergence of Markov Chain Monte Carlo methods: an example from evaluation of diagnostic tests in absence of a gold standard. *Prev. Vet. Med.* 79, 244–256.
- Toft, N., Åkerstedt, J., Tharaldsen, J., Hopp, P., 2007b. Evaluation of three serological tests for diagnosis of Maedi-Visna virus infection using latent class analysis. *Vet. Microbiol.* 120, 77–86.
- Vacek, P.M., 1985. The effect of conditional dependence on the evaluation of diagnostic tests. *Biometrics* 41, 959–968.